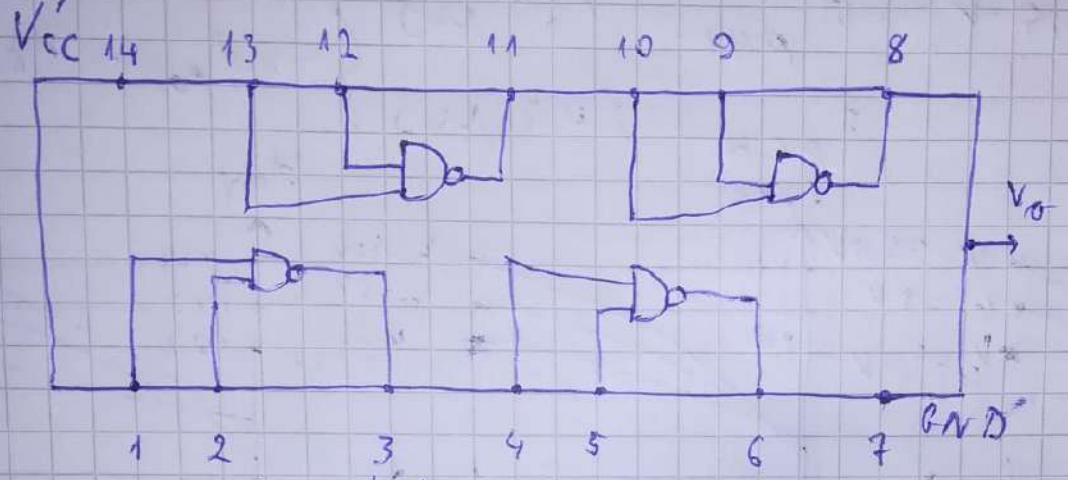
Date experimentale CI pentru:

1) <u>Aplicația 1</u>:

- poarta TTL 400



1,6 V 0 V

(= „1" logic) ⇒ („0" logic)

- circuitul cu o diodă și R_L

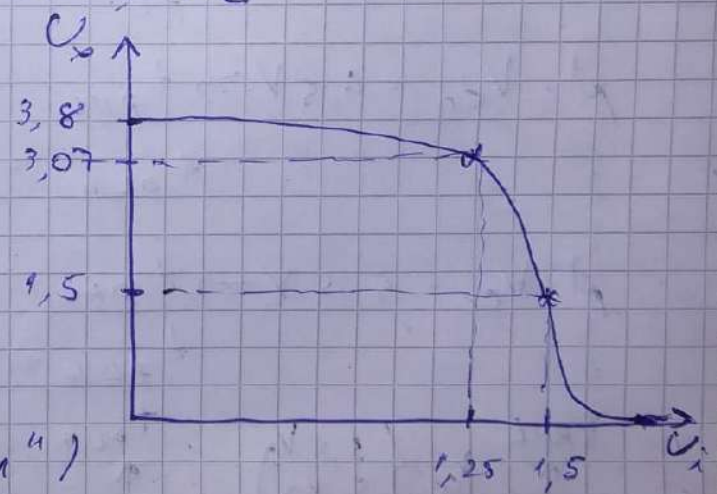
$R_L = 3,9\ k\Omega$

$V_{CC} = 5\ V$

Channel 2 – U_i

Channel 1 – U_o

U_i	U_o
0	3,8 V → V_{IH} („1")
0,25	3,8 V
0,5	3,8 V
0,75	3,7 V
1	3,47 V
1,25	3,07 V
1,5	1,5 V



2) Aplicația 2:

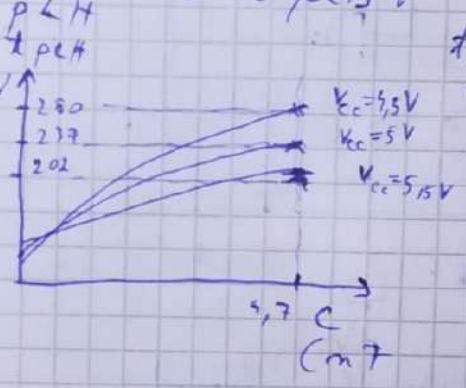
a) $C_1 = 4,7\ mF$ $\quad T = 100\ \mu s$

b) $C_2 = 220\ pF$

a) pt. $V_{CC} = 5V \Rightarrow \left.\begin{array}{l} t_{pHL} = 149\ \mu s \\ t_{pLH} = 237\ \mu s \end{array}\right\} \Rightarrow t_{pd} = 193\ \mu s$

pt. $V_{CC} = 4,5V \Rightarrow \left.\begin{array}{l} t_{pHL} = 145\ \mu s \\ t_{pLH} = 280\ \mu s \end{array}\right\} \Rightarrow t_{pd} = 192,5\ \mu s.$

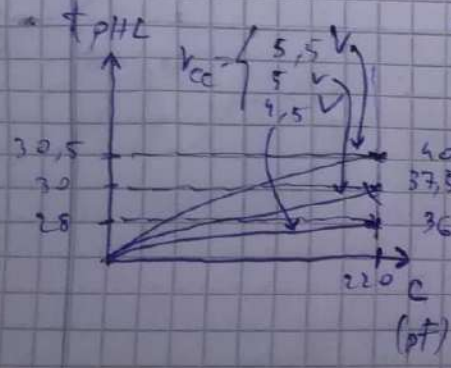
pt. $V_{CC} = 5,5V \Rightarrow \left.\begin{array}{l} t_{pHL} = 162\ \mu s \\ t_{pLH} = 202\ \mu s \end{array}\right\} \Rightarrow t_{pd} = 182\ \mu s.$

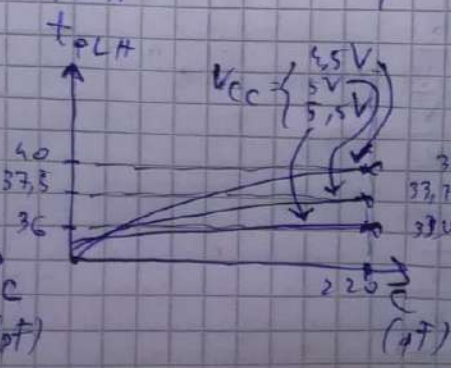


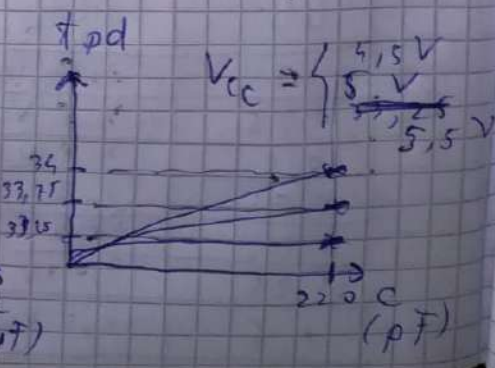
b) pt. $V_{CC} = 5V \Rightarrow \left.\begin{array}{l} t_{pHL} = 30\ ms \\ t_{pLH} = 37,5\ ms \end{array}\right\} \Rightarrow t_{pd} = 33,75\ ms$

pt. $V_{CC} = 4,5V \Rightarrow \left.\begin{array}{l} t_{pHL} = 28\ ms \\ t_{pLH} = 40\ ms \end{array}\right\} \Rightarrow t_{pd} = 34\ ms$

pt. $V_{CC} = 5,5V \Rightarrow \left.\begin{array}{l} t_{pHL} = 30,5\ ms \\ t_{pLH} = 36\ ms \end{array}\right\} \Rightarrow t_{pd} = 33,25\ ms.$







3) <u>Aplicatia 3:</u> $R = 390\,\Omega$; $R \uparrow \Rightarrow T \uparrow$

$T_1 = t_2 - t_1 = 6\ \mu s$

$T_2 = t_3 - t_2 = 4\ \mu s$

De pe cursor: $T_1 = 5{,}43\ \mu s$

$T_2 = 4{,}17\ \mu s$.

4) <u>Aplicatia 4:</u>

V_x: 1,32 V offset

$\Delta V = 1{,}66$ V amplitudine

V_y: 1,08 V offset

$\Delta V = 2{,}4$ V amplitudine

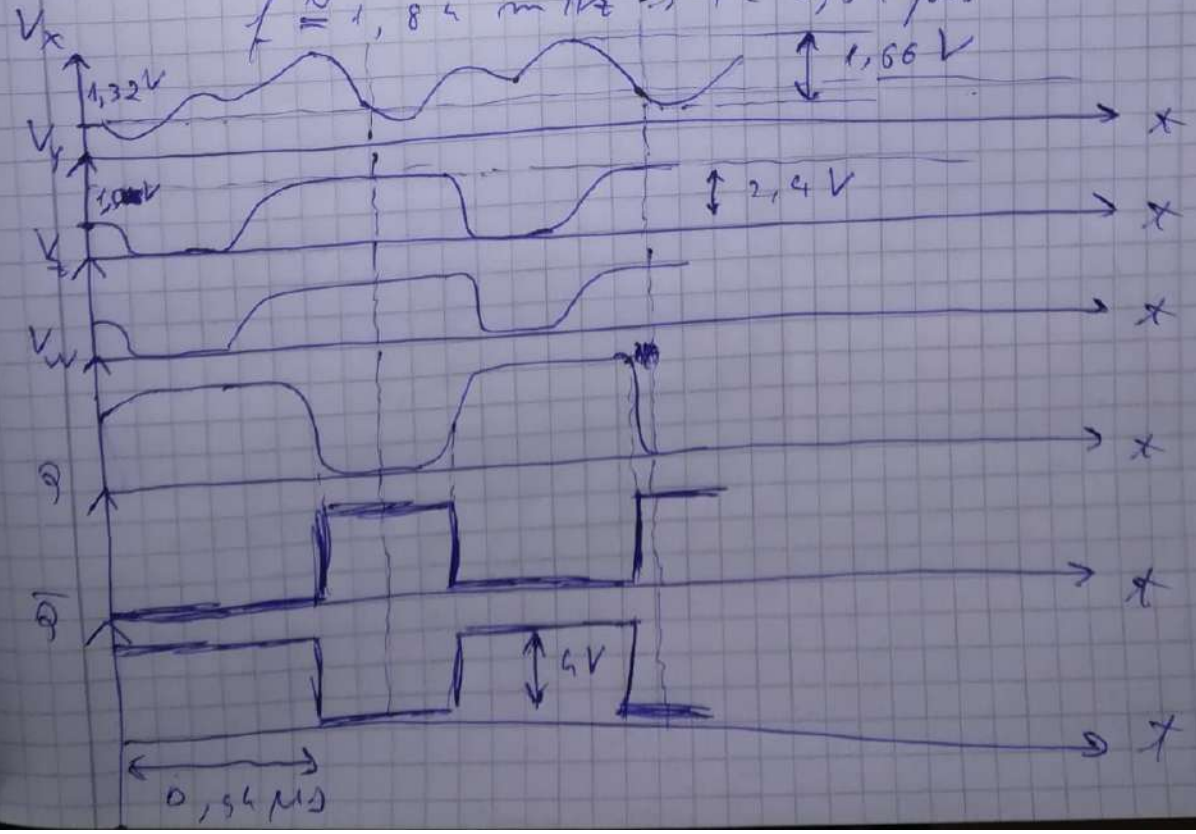
V_z: 1,1 V offset

$\Delta V = 2{,}39$ V amplitudine

V_w: V_y

$Q, \overline{Q} \rightarrow A \cong 4$ V

$f \cong 1{,}84$ MHz $\Rightarrow T = 0{,}54\ \mu s$



5) Aplicatia 5:

a) $R = 330\,\Omega$

$C = 10^5\ pF \simeq 100\,nF$

$T_{i\,măsurat} = 39\ \mu s$

$T_i = 330 \cdot 100 \cdot 10^{-9} \cdot \ln \frac{0,2-4}{0,2-1,5} = 33 \cdot 10^{-6} =$

$= 33\ \mu s$

c) $R = 330\,\Omega$

$C = 10^5\ pF$

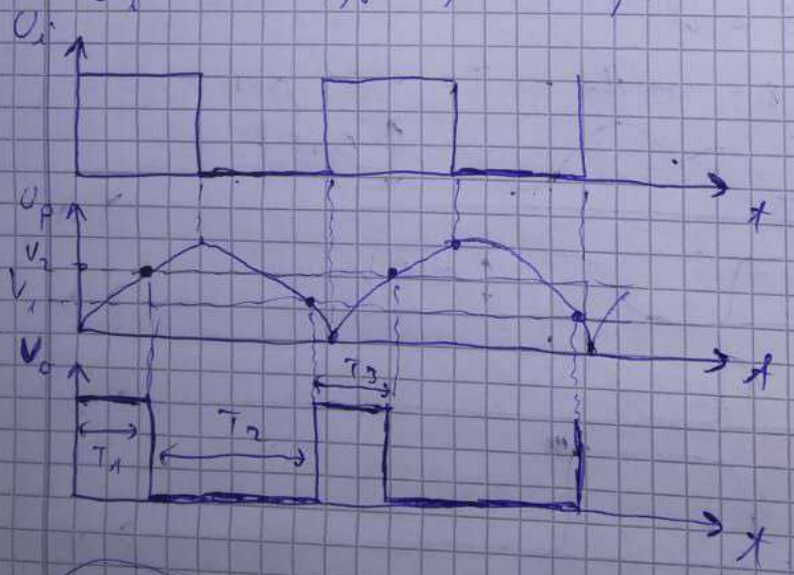
$T_{i\,măsurat} = 43\ \mu s$

$T_i = RC \ln \frac{0,2-4}{0,2-1,5} = 33\ \mu s$

6) Aplicatia 6:

9.3 a

U_i : $A = 3,5\ V$; $T = 2\ \mu s$



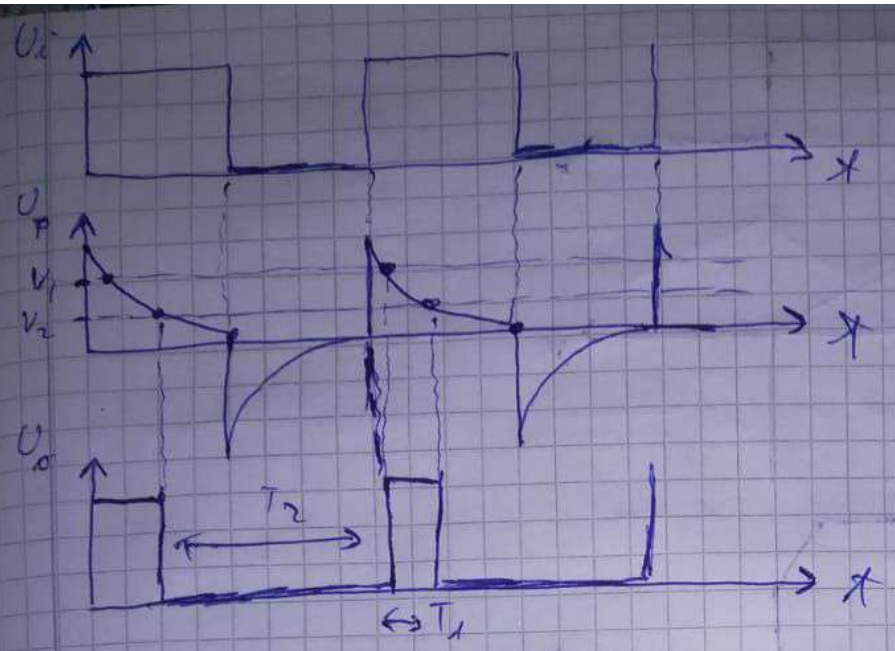
$T_1 = 194\ ns$

$T_2 = 1,36\ \mu s$

$T_3 = 720\ ns$

9.4 b

U_i : $A = 3,5\ V$; $T = 50\ \mu s$



$T_1 = 360\ ms$

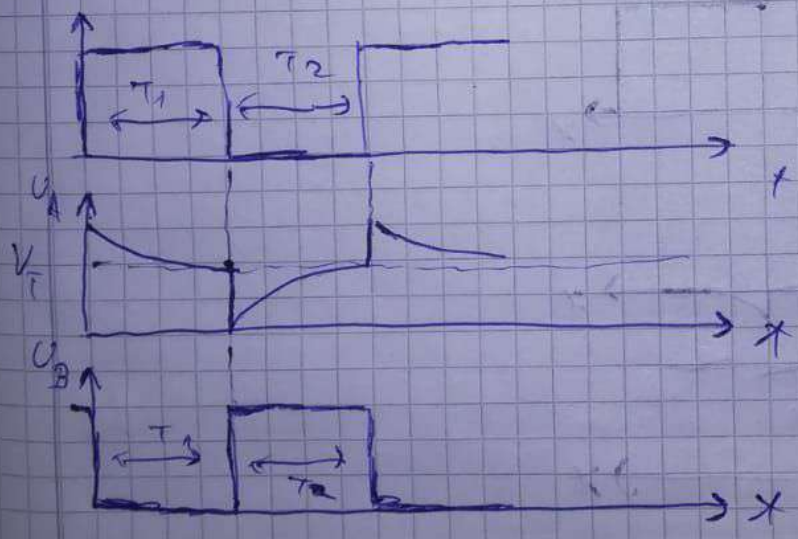
$T_2 = 48,6\ \mu s$

7) Aplicatii:

A.4.1

$V_{DD} = 5V$; $R = 330\ k\Omega$; $C = 1\ nF$ $R = 370\ k\Omega$

$V_{DD} = 12V$



$T_1 = 291\ \mu s$

$T_2 = 201\ \mu s$

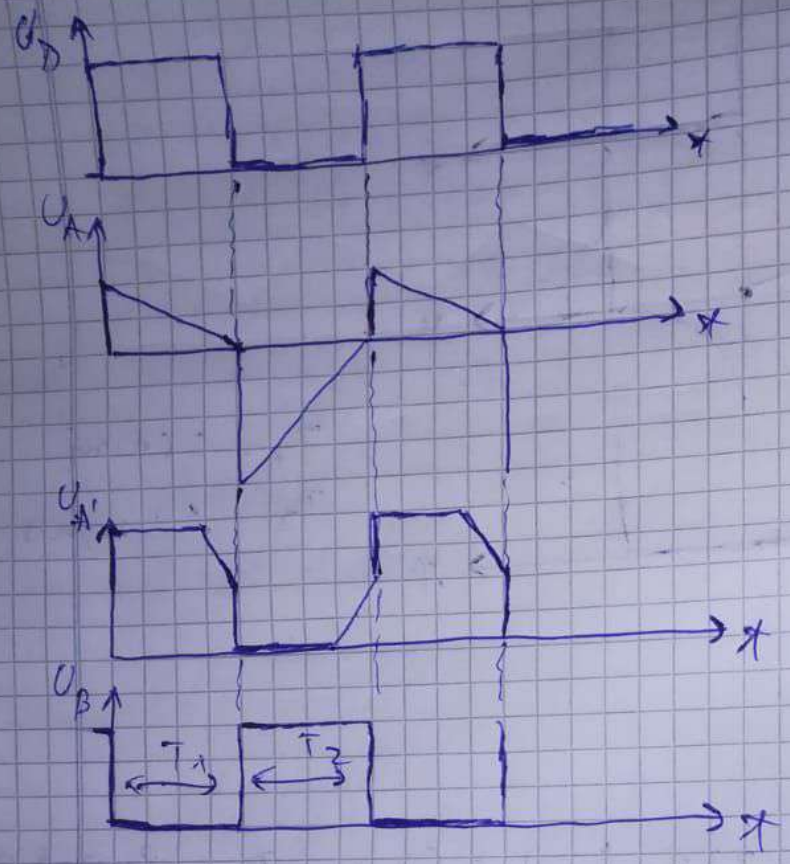
A.4.3 $R_1 = 320\ k\Omega$, $C = 1\ nF$; $V_{DD} = 5V$;

$R_2 = 750\ k\Omega$

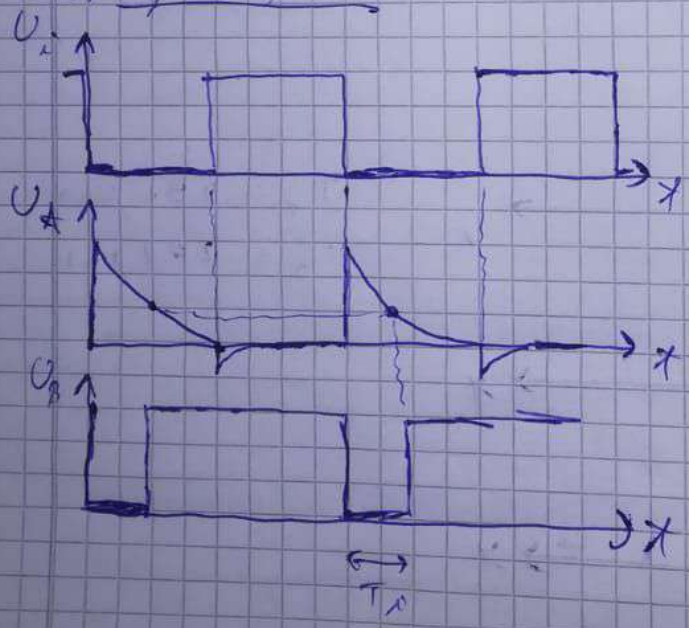
$T = 322\ \mu s$

$T_1 = 247\ \mu s$

$T_2 = 247\ \mu s$



8) Aplicația 8: $T = 50\,\mu s$; $f = 26\,kHz$

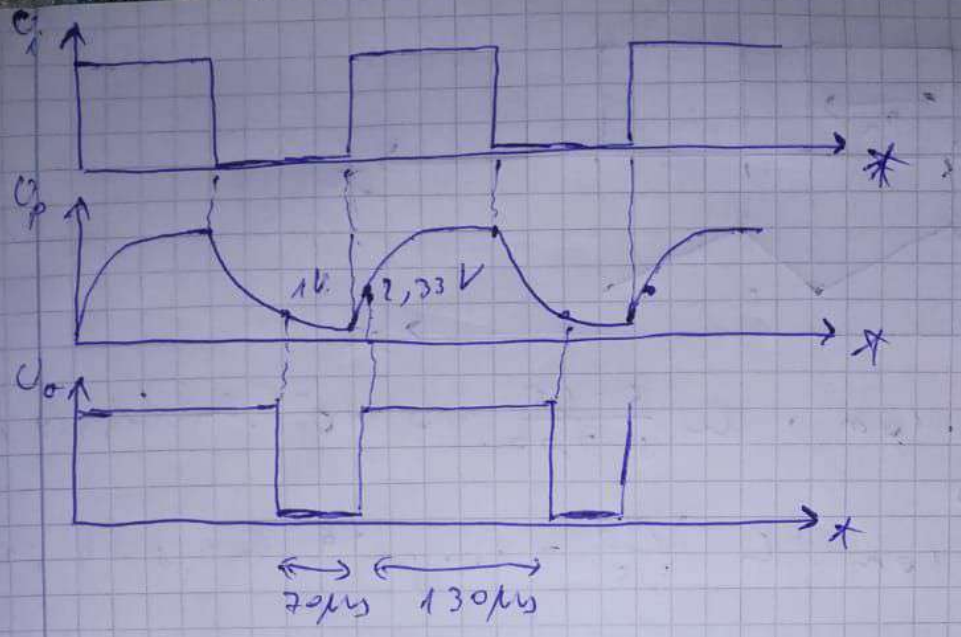


$$T_1 = RC \ln \frac{V_{DD}}{V_T} = 13k \cdot 1n \ln \frac{5}{2,5} = 9\,\mu s$$

$$T_1 \cong 12\,\mu s$$

$U_2 = 1\,V$

$U_1 = 2,33\,V$



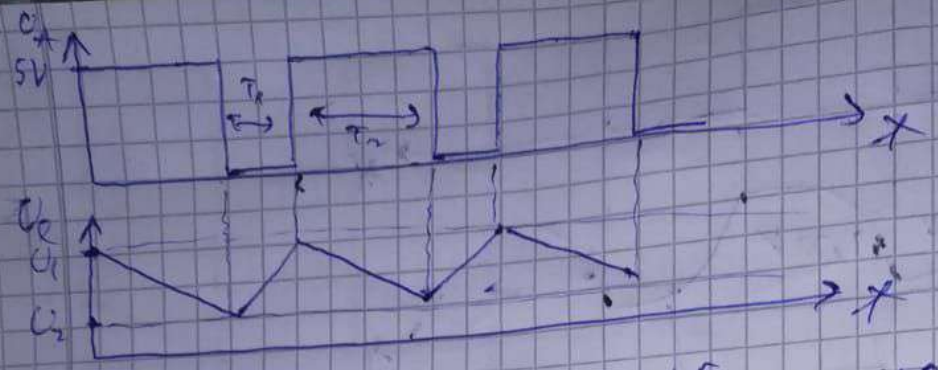
U_i	U_o
0V	0V
0,25 V	0V
0,5 V	0V
0,75 V	0V
1 V	0V
1,25 V	0V
1,5 V	0V
1,75 V	0V
2 V	0V
2,25 V	0V
2,5 V	0V
2,6 V	5V
⋮	⋮
5 V	5V

U_i	U_o
5 V	5V
⋮	⋮
2 V	5V
1,75 V	5V
1,65 V	0V
⋮	⋮
0V	0V

9) Aplicația 9:

~~A 2.1 afișați: f = 5 KHz~~
~~pripadă U_i T = 4,50μs 200μs~~

A 2.4 U_C; U_A; T_1; T_2



$T_1 = 1k \cdot 10n \cdot \ln \frac{5-1,5}{5-2,77} = 10 \cdot 10^{-6} \ln (1,569)$

$= 4,5 \mu s$

$T_2 = 1k \cdot 10n \cdot \ln \frac{2,77}{1,5} = 10 \cdot 10^{-6} \cdot \ln 1,8 = 5,87 \mu s$

10) Aplicatia 11:

A.33.

$U_p = 3V$

$V_N = 2V$

$V_{DD} = 5V$

$\Delta t_1 = 10 \cdot 10^3 \cdot 2,2 \cdot 10^{-9} \cdot 0,91629 = 20,15 \mu s$

(teoretic)

$V_N = 2,3V$

$V_p = 3,10V$

$C = 1,7n$

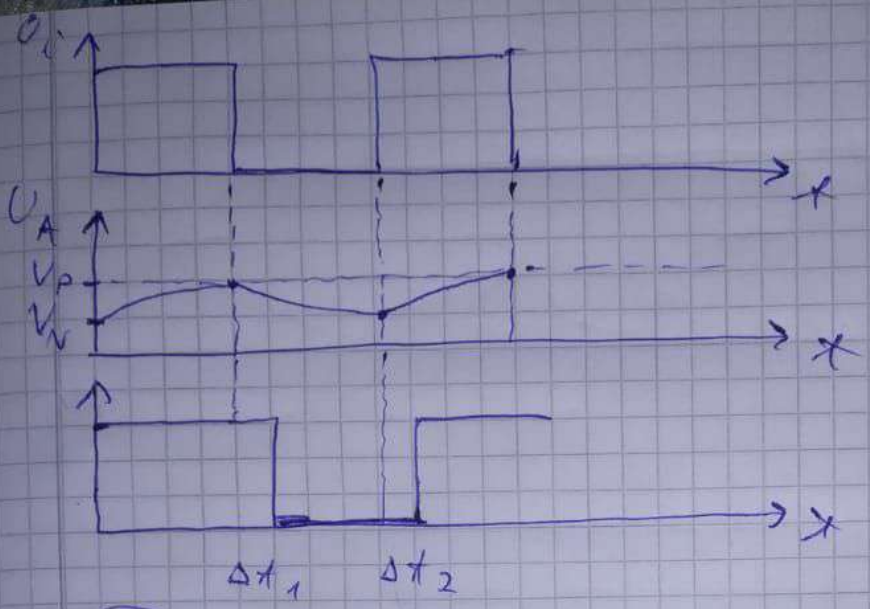
$\Delta t_1 = 10 \cdot 10^3 \cdot 1,7 \cdot 10^{-9} \cdot \ln \frac{4,8 - 0,7}{1,89} = 17 \cdot 10^{-6} \cdot$

$0,667 = 11,353 \mu s$

$\Delta t_1 \approx 8,63 \mu s$

$\Delta t_2 = 24 \cdot 10^3 \cdot 2,2 \cdot 10^{-9} \cdot \ln \frac{0,7 - 4,3}{0,7 - 2,3} = 42 \mu s$

(practic)



A. 3.4

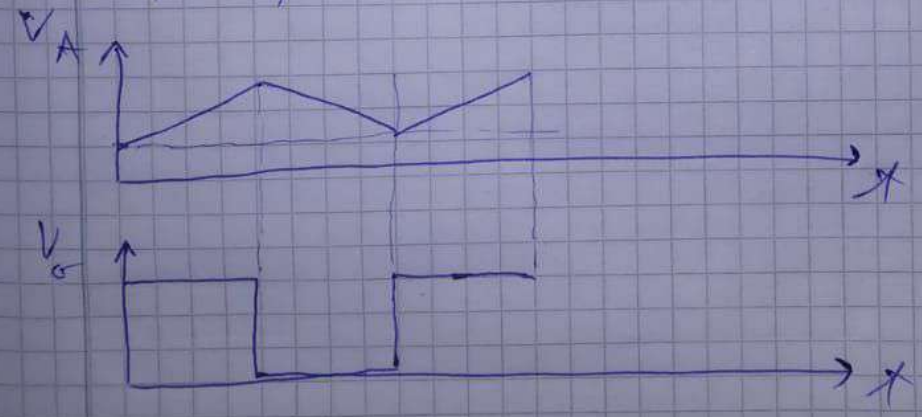
(teoretic) $V_N = 2,3V$; $V_P = 3,16V$

$T_1 = 10 \cdot 10^3 \cdot 100 \cdot 10^{-12} \cdot \ln \frac{4,3-2,3}{4,3-3,16} = 1000 \cdot 10^{-9} \cdot$

$\ln \frac{2}{1,14} = 0,56 \mu s.$

(practic) $T_1 = 3,28 \mu s$

(teoretic)

$T_2 = 10 \cdot 10^3 \cdot 100 \cdot 10^{-12} \cdot \ln \frac{2,3}{3,16} = 10^{-6} \cdot 0,317 =$

$= 0,317 \mu s$